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Pharmacology: Parkinson’s and Alzheimer’s disease

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Session Objectives

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1. Review the pathogenesis of Parkinson’s disease.
2. Describe all the classes of drugs used in the management of Parkinson’s disease. Identify adverse effects
3. Explain the drugs which can induce Parkinsonism.
4. Review the pathogenesis of Alzheimer’s disease
5. Describe the drugs used to manage Alzheimer’s disease. Include novel therapies.

Source: Course Syllabus

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Parkinson's Disease

- Loss of dopaminergic neurons in substantia nigra.
- Rigidity, postural instability, bradykinesia, tremor, affective disorders.

Drugs with PD-like effects:

- Antipsychotics (D_2 receptor blockers; haloperidol, fluphenazine, chlorpromazine)
- Antiemetics (metoclopramide, prochlorperazine)
- Vesicular monamine transporter-2 inhibitor (tetrabenazine). Depletes monoamines, used for chorea.

Dopamine Pathway

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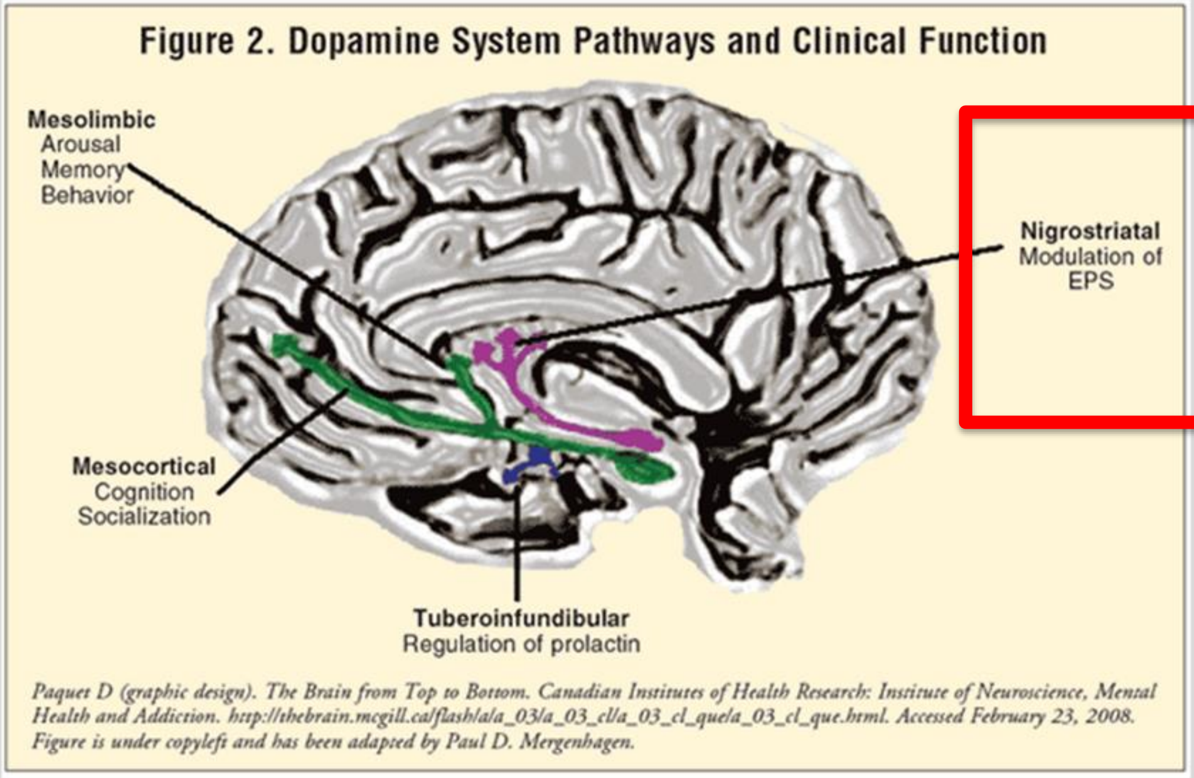

Parkinson's Disease Pathogenesis

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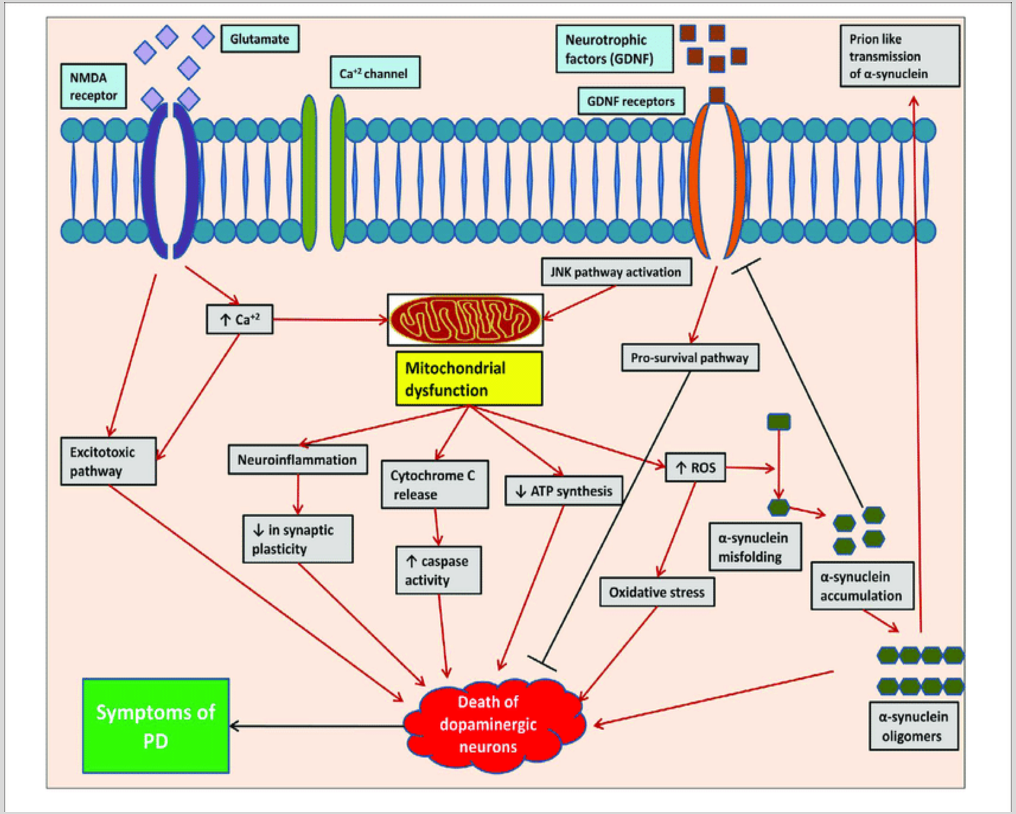
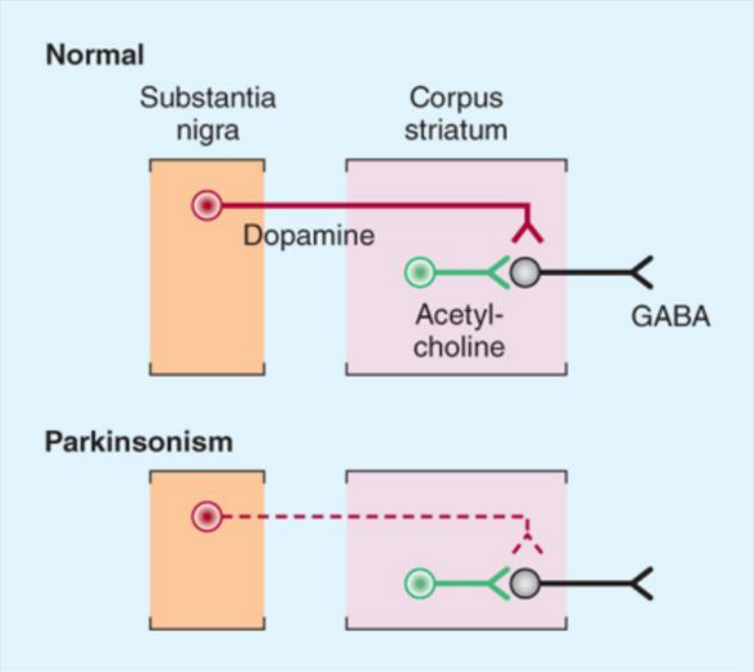


https://www.researchgate.net/figure/Pathogenesis-of-Parkinsons-disease-at-molecular-level-showing-different-complex_fig2_324803436



L-Dopa

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- Precursor of Dopamine (DA). L-dopa crosses BBB
- Converted to dopamine by dopa decarboxylase in nigrostratium
- Drug efficacy ceases as dopaminergic neurons are lost
- Dyskinesia (involuntary movements). Relates to unequal distribution of striatal dopamine.
- On-off phenomena. Drug holiday not effective. Use other drugs.

L-Dopa Metabolism

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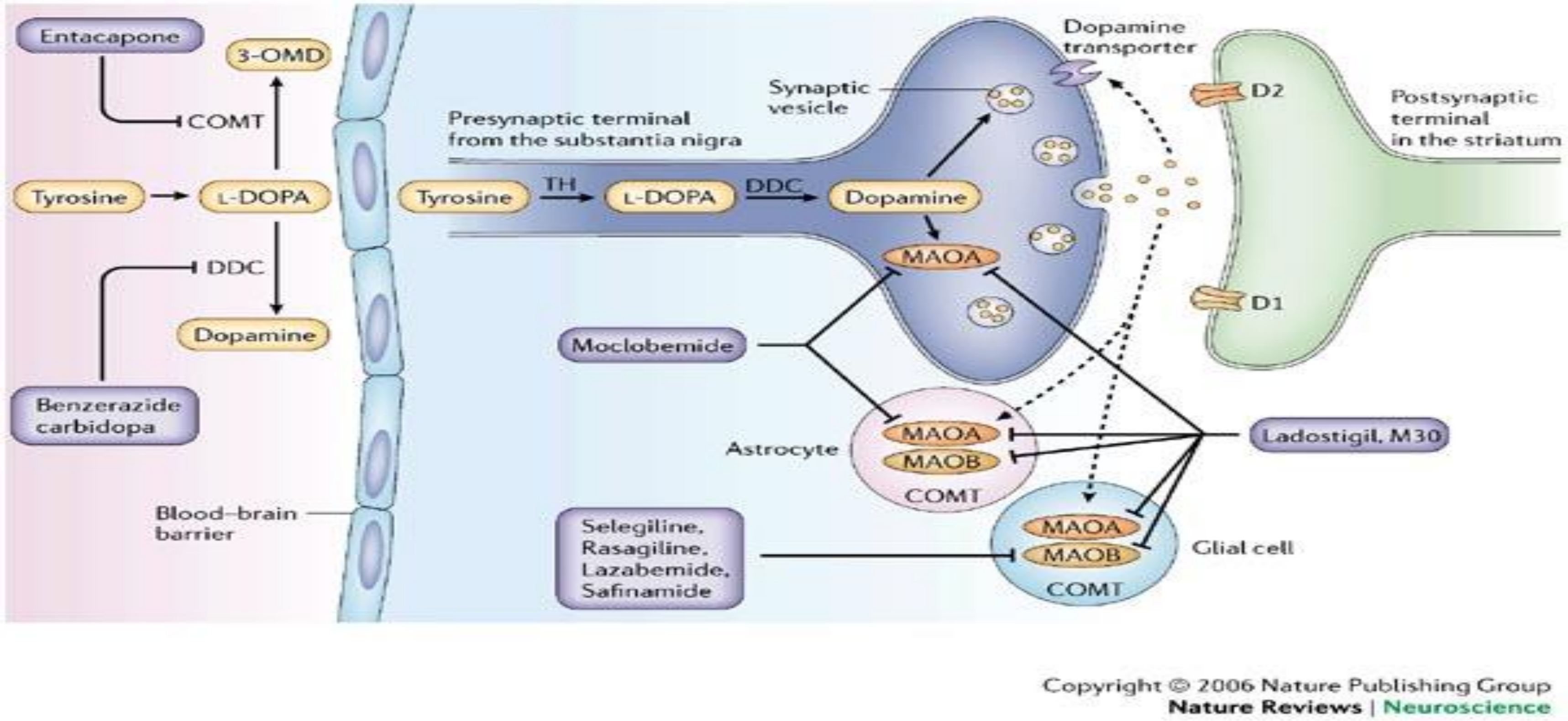

Nature Reviews | Neuroscience

Youdim *et al.* *Nature Reviews Neuroscience* **7**, 295–309 (April 2006) | doi:10.1038/nrn1883

L-Dopa Adverse effects

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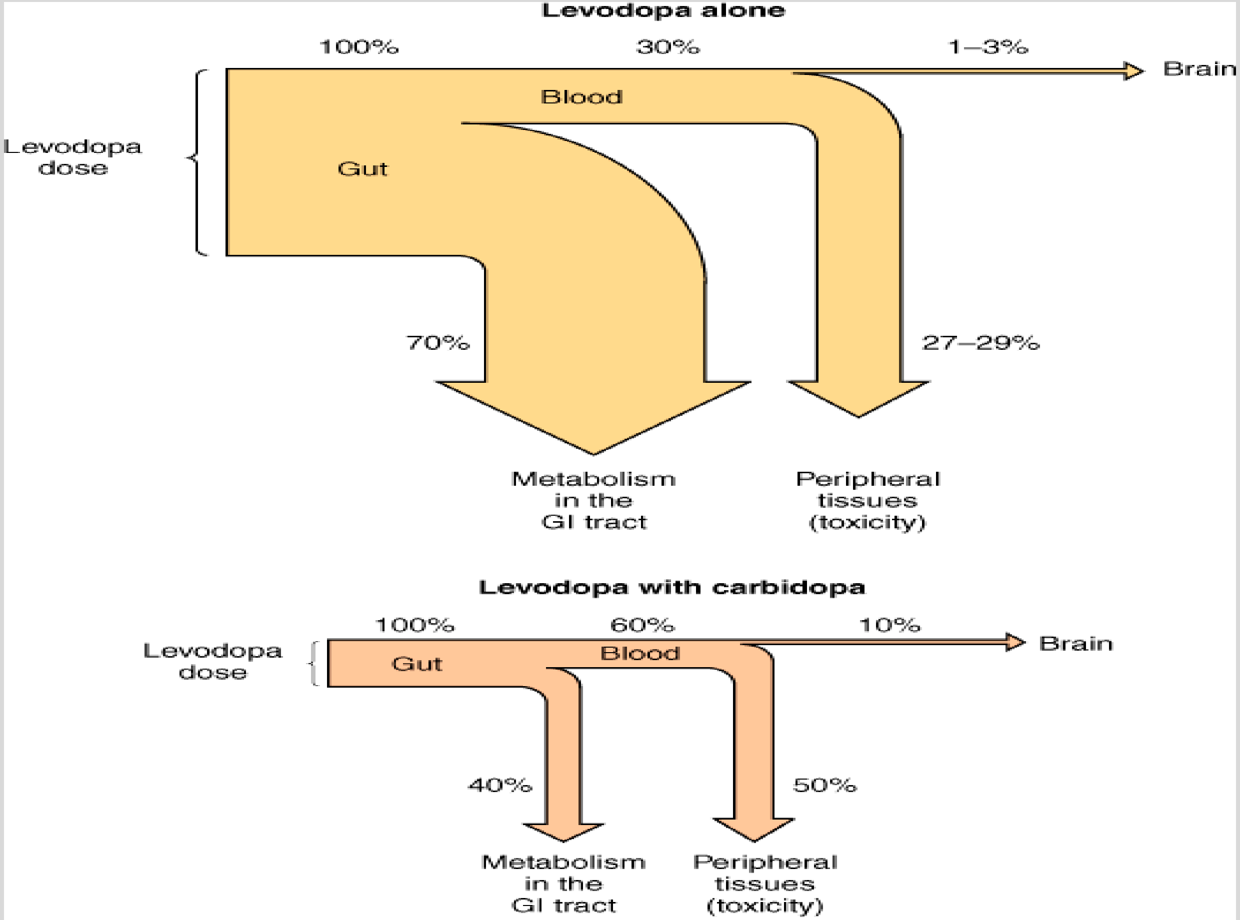
- **Nausea, vomiting (DA in CTZ)**
- **Orthostatic hypotension**
- **Cardiac arrhythmia**
- **Hallucinations, delusions, compulsive behavior. May be relieved by atypical antipsychotics.**
- **Carbidopa decreases adverse effects (dopa decarboxylase inhibitor). Reduces L-dopa dose requirement. Peripheral action.**



Source: Katzung BG, Masters SB, Trevor AJ: *Basic & Clinical Pharmacology, 11th Edition*: http://www.accessmedicine.com

L-dopa interactions

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- High protein foods decrease L-dopa absorption.
- Vitamin B_6 increases decarboxylation
- MAO-A inhibitors (can induce hypertensive crisis with L-dopa)
- Metabolites: homovanillic acid; dihydoxyphenylacetic acid (DOPAC)

Amantadine

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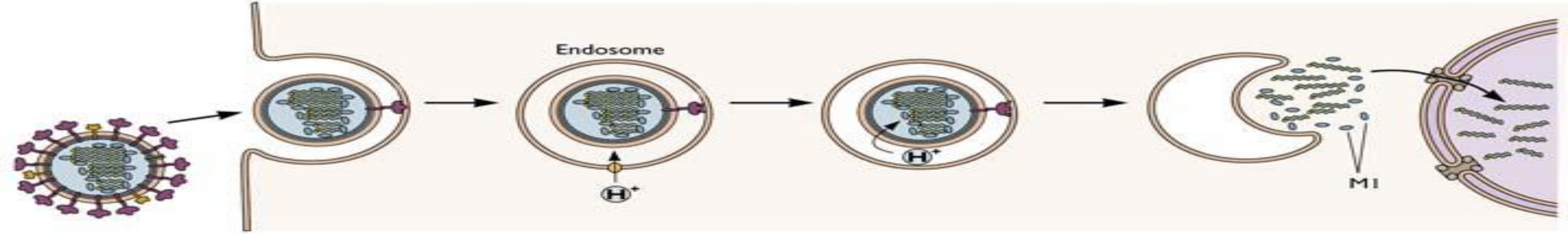
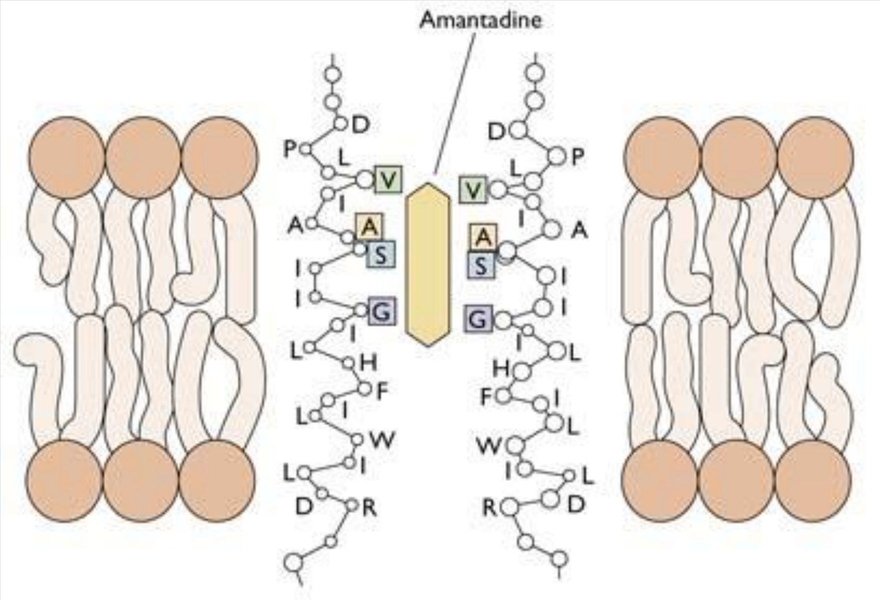
- Treatment and prophylactic (influenza A)
- For PD: Increases dopamine release, muscarinic blockade, and NMDA receptor block (antidyskinetic effect).
- Oral, Renal clearance
- Adverse effects: Dopaminergic, muscarinic block effects, livedo reticularis (skin)



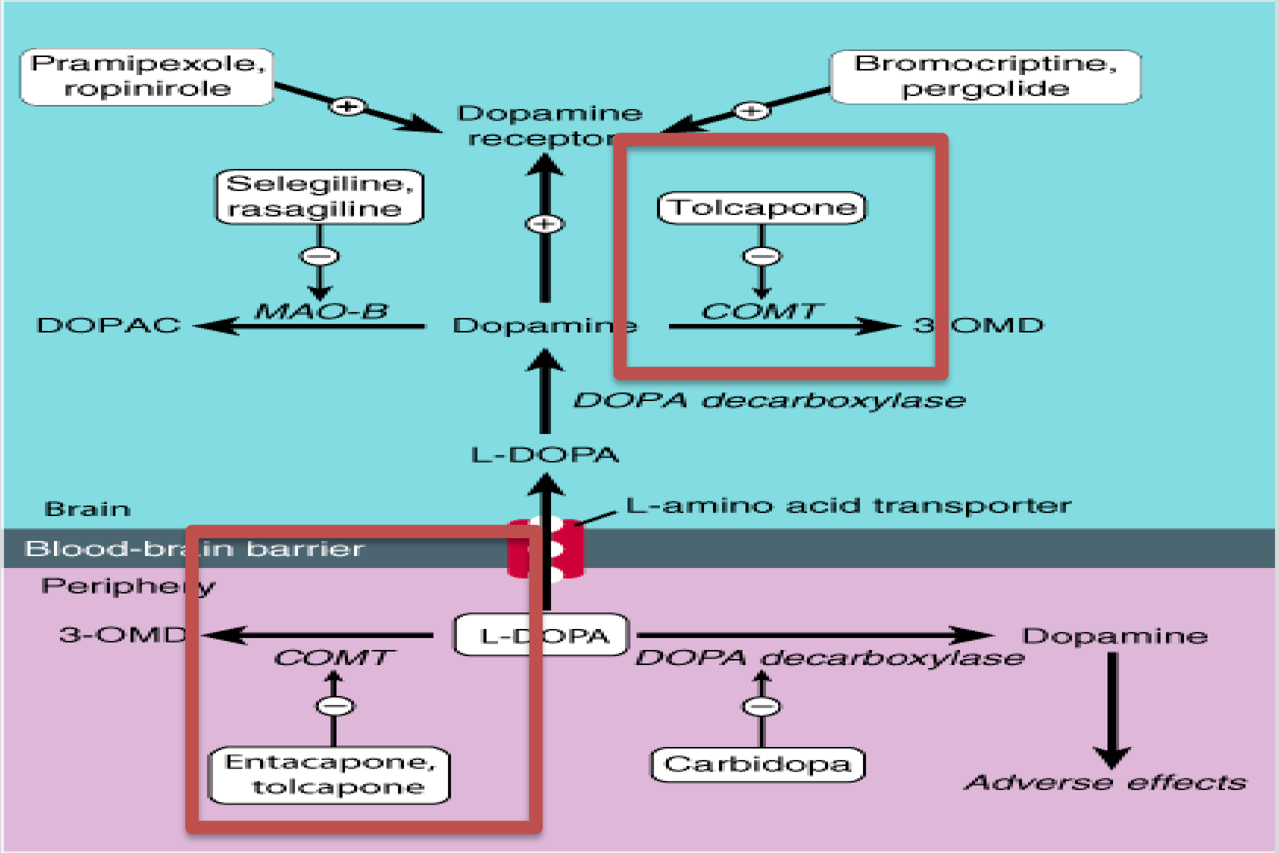
H+ must enter virus via M_2 ionic channel to acidify pH and cause uncoating and release of RNA. Amantadine blocks M_2 channel, prevents release of RNA. This is antiviral MOA.

Resistance: Mutation of the M_2 protein



Catechol-O-Methyltransferase Inhibitors

- Tolacapone and Entacapone
- Stabilizes dopamine levels
- Increases bioavailability of levodopa
- Tolacapone: Acute hepatitis
- Entacapone: safer, no liver toxicity. But only acts in periphery
- Opicapone (long acting)
- Adverse effects: related to increase in L-dopa levels (N/V, cardiac, dyskinesias).



Source: Katzung BG, Masters SB, Trevor AJ: *Basic & Clinical Pharmacology, 11th Edition*: http://www.accessmedicine.com

Levodopa, Carbidopa, Entacapone

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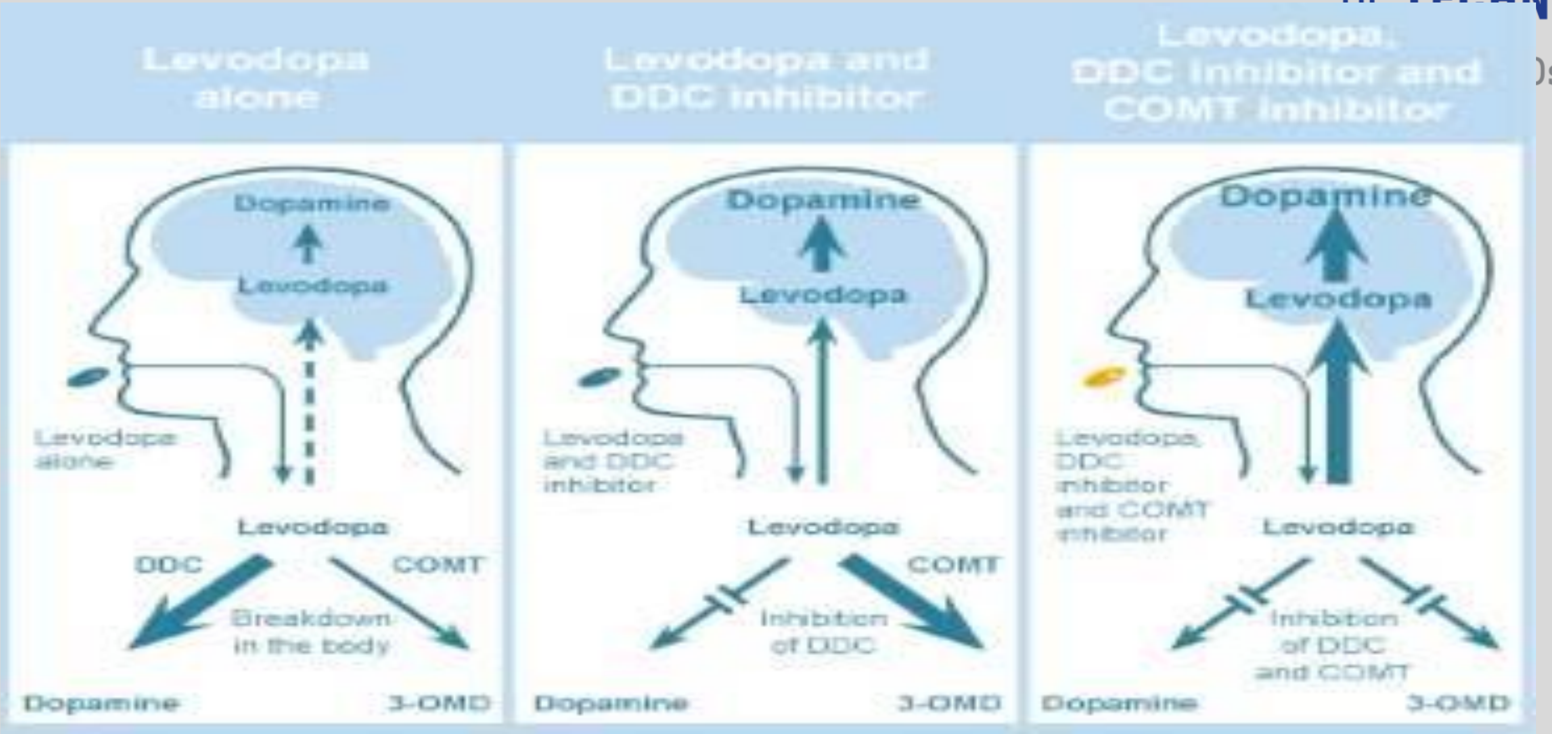
Osteopathic



http://www.orion.fi/en/Research-and-developement/Research-Areas-/Central-Nervous-System-medicines/

MAO-B Inhibitors

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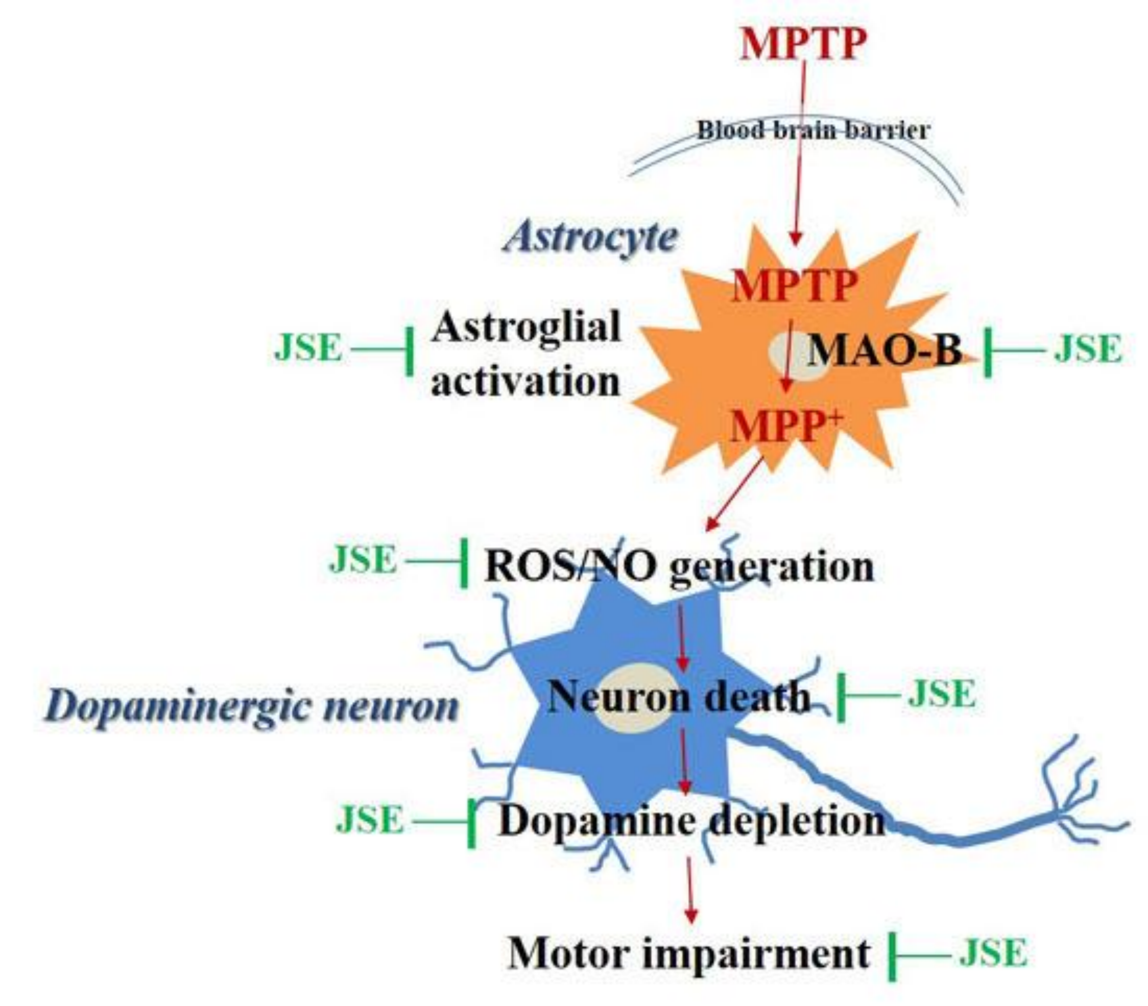
- Selegiline, Rasagiline (irreversible).
- Selegiline metabolite is metamphetamine
- Safinamide (decreases off phenomena, reversible). Inhibits glutamate release.
- MAO-B blockade:
  - Increases DA levels, allows for levodopa dose reduction.
  - Decreases hydrogen peroxide formation (less ROS)
  - Prevents MPTP induced Parkinsonism (toxin)



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https://www.mdpi.com/1422-0067/17/1/108/htm

MAO-B Inhibitors: Adverse effects

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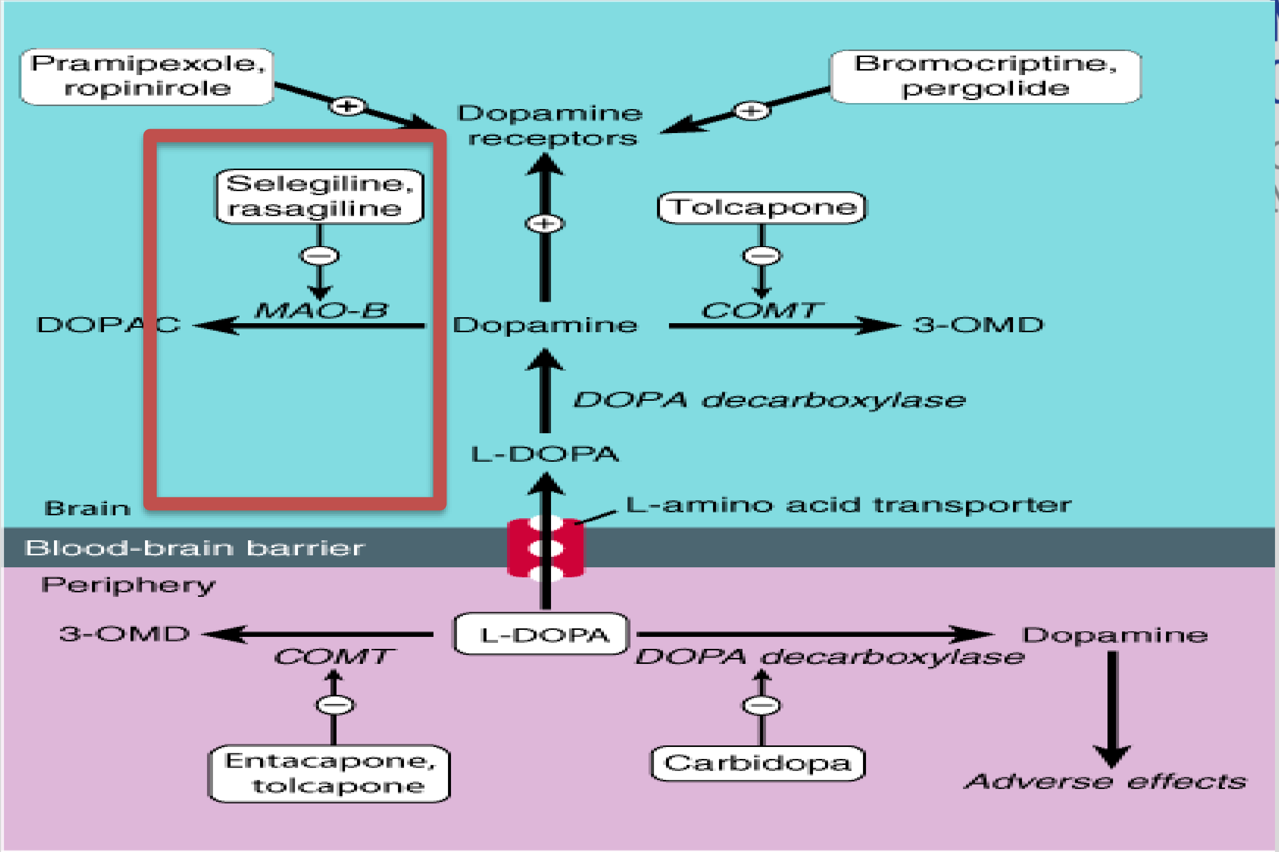
- MAO-B: dopamine specific. Minimal interaction with tyramine-containing food (unlike phenelzine). Unless administered at high doses.
    - Selegiline metabolized into amphetamine.
    - Serotonin syndrome (with co-administration of meperidine, dextromethorphan, St. John's wort)
    - Safinamide not indicated in severe hepatic impairment.
    - MAO-A: NE, 5HT, dopamine (phenelzine). Antidepressants



Source: Katzung BG, Masters SB, Trevor AJ: *Basic & Clinical Pharmacology, 11th Edition*: http://www.accessmedicine.com

D_2 and D_3 Receptor Agonists

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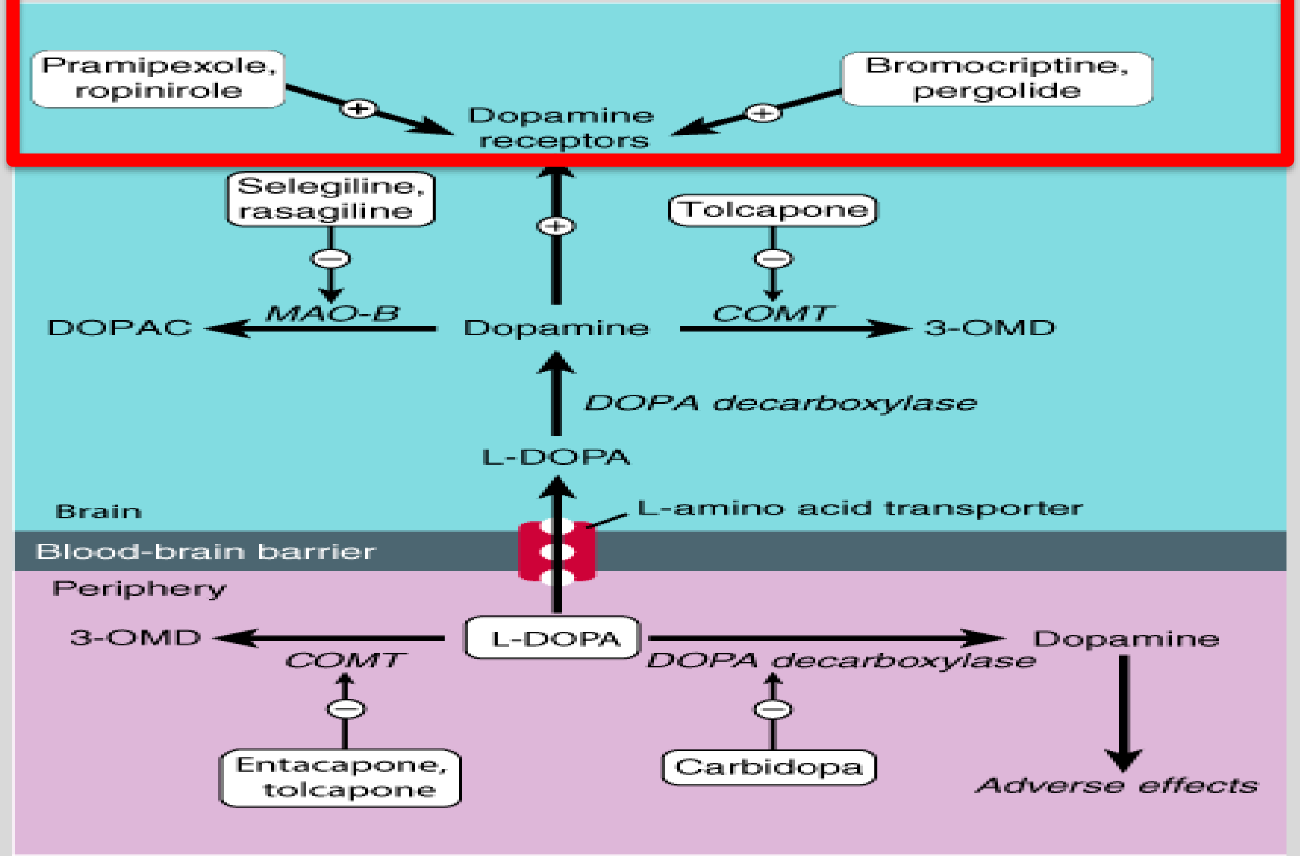
- Pramipexole: D_3 receptor agonist. Scavenger of hydrogen peroxide.
- Ropinirole: D_2, D_3 receptor agonist
- Bromocriptine: D_2 receptor agonist. Treats elevated prolactin. Ergot derivative (vasospasm)
- Rotigotine: transdermal formulation
- Administer with L-dopa; reduces on-off phenomena
- Adverse effects: confusion, vivid dreams, hallucinations, N/V, compulsive behavior (D_3).
- Also used for restless legs syndrome (pramipexole, ropinirole)



Source: Katzung BG, Masters SB, Trevor AJ: *Basic & Clinical Pharmacology, 11th Edition*: http://www.accessmedicine.com

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Apomorphine

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- **Dopamine receptor agonist (nonergoline). Postsynaptic D_2 in caudate nucleus and putamen.**
- **Injectable, sq, sl to provide supplemental dopamine action.**
- **Limits "off" episodes**
- **<u>Adverse effects</u>: Nausea and vomiting. Can be managed by trimethobenzamide.**
- **Increased QT interval.**

Balance Hypothesis of Striatal Function

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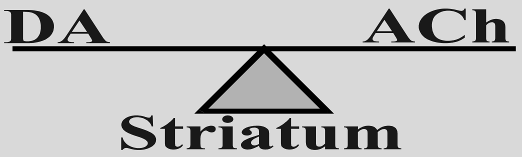

↔ DA & ↔ ACh = Normal Motor Behavior

↓ DA & ↔ ACh =
↔ DA & ↑ ACh =
Hypokinesia (Parkinson's disease and dystonia)

↑ DA & ↔ ACh =
↔ DA & ↓ ACh =
Hyperkinesia (Choreatic disorders, Tourette's syndrome, hyperactivity disorders, and dyskinesias)

Figure 9-6

Muscarinic Antagonists

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- Benztropine, trihexyphenidyl
- Restore some of the “balance”(Ach vs DA)
- Reduce tremor and rigidity
- Adverse effects: Antimuscarinic
- Elderly????
  - M_1: CNS, M_2: cardiac; M_3: exocrine glands

5-HT$_{2A}$, Alpha$_1$, M$_3$, H$_1$ Mechanism of Action (Gq)

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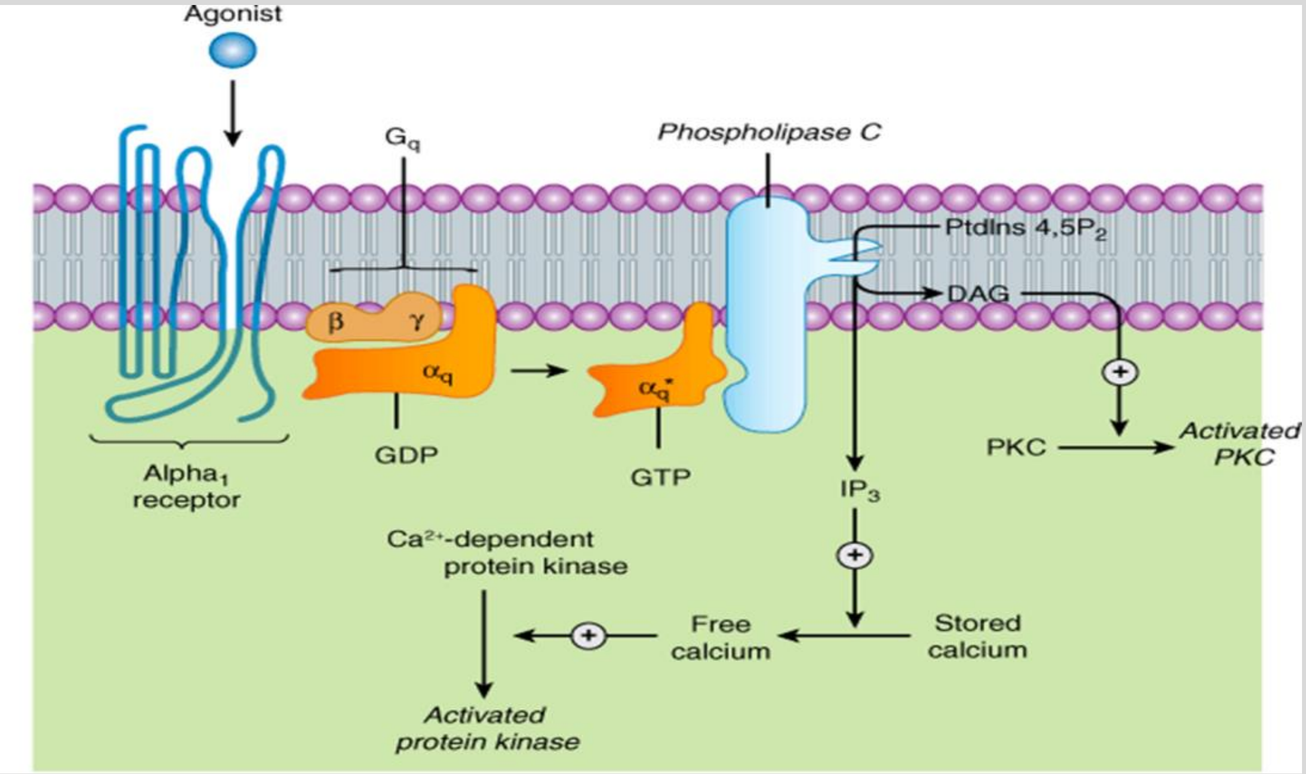

Istradefylline

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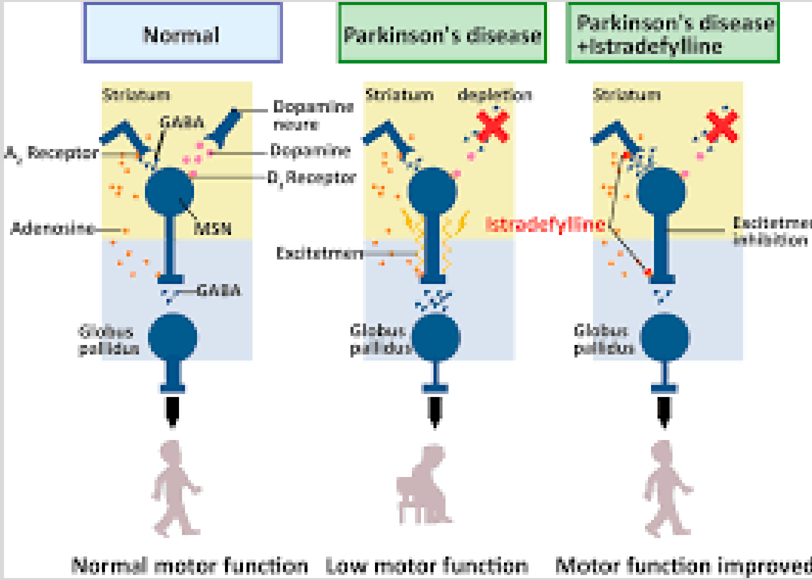
- Selective adenosine A_{2A} receptor antagonist in basal ganglia
- GABAergic neurotransmission is reduced in globus pallidus
- Manages the off-phenomena
- <u>Adverse effects</u>: insomnia, reduced appetite



Alzheimer’s disease

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- Most common form of dementia, progressive
- Loss of brain cells to impair memory & cognition.
- Greater incidence with age, 20% in patients over 85.
- Familial & sporadic (single family member) forms identified. APOE-4 gene
- No cure, only slow progression.

Pathology

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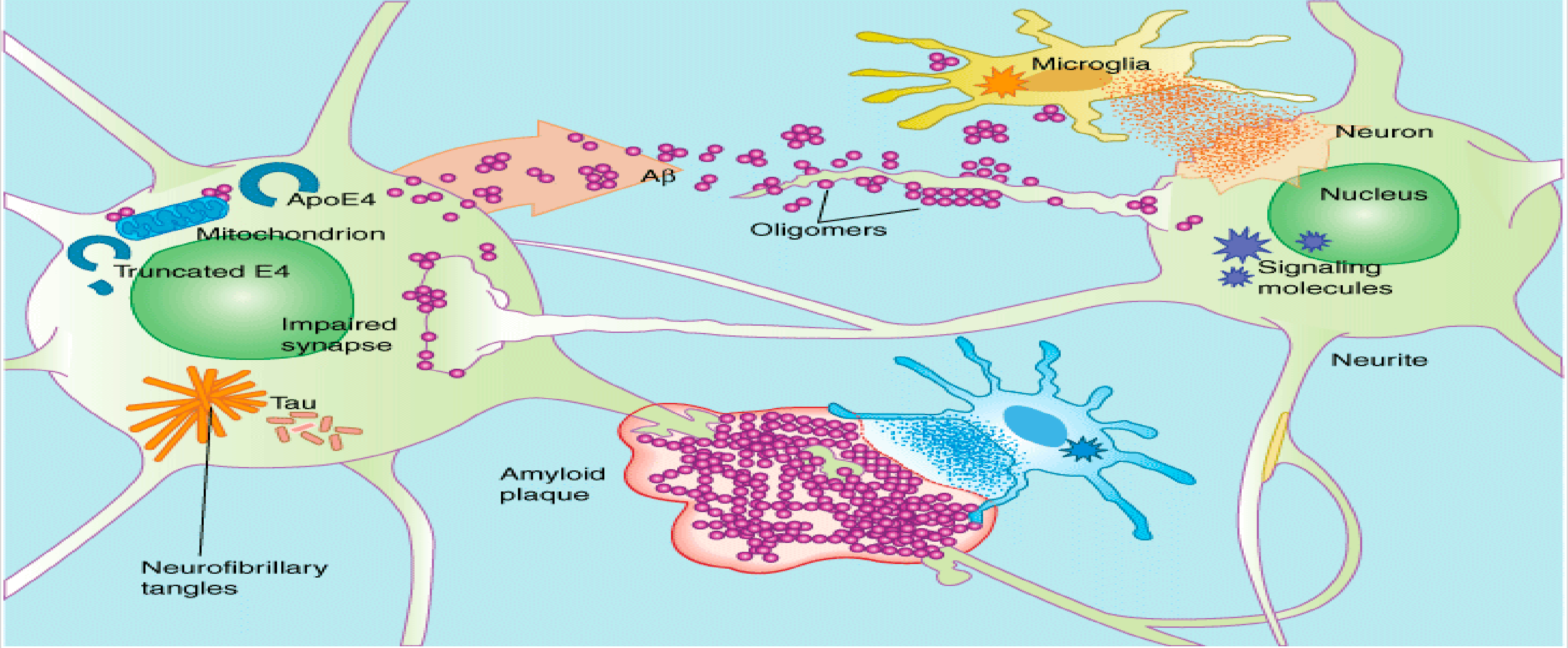
- Intraneuronal fibrillary tangles consisting of the tau protein.
- Amyloid-β deposited in the cerebral cortex, cerebrovascular lesions.
- Progressive loss of neurons in cerebral cortex and cholinergic neurons.
- Decreased acetylcholine synthesis.



Source: Katzung BG, Masters SB, Trevor AJ: *Basic & Clinical Pharmacology, 11th Edition*: http://www.accessmedicine.com

From the left: mitochondrial dysfunction, possibly involving glucose utilization; synthesis of protein tau & aggregation in filamentous tangles; synthesis of amyloid β (Aβ) & secretion into the extracellular space, where it may interfere with synaptic signaling and accumulates in plaques.

Anticholinesterase Inhibitors

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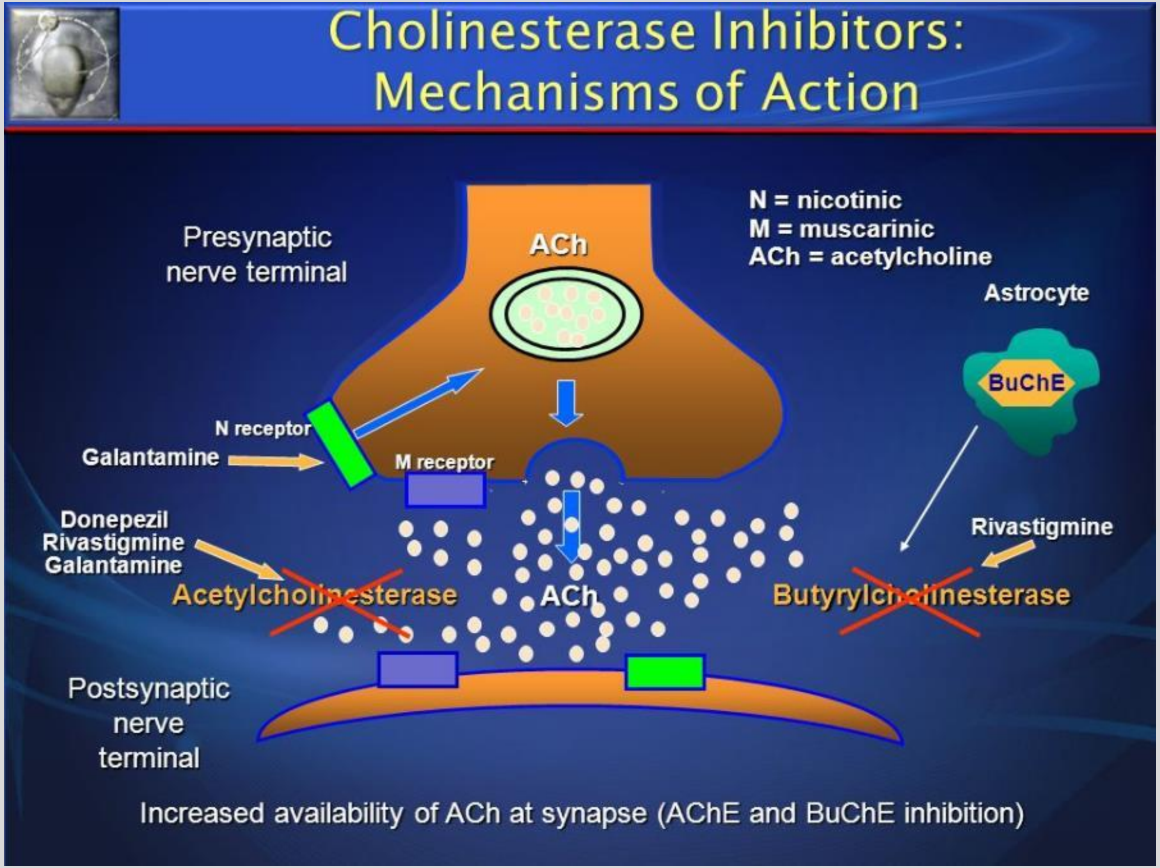
- Donepezil, Rivastigmine, Galantamine, Tacrine
- Approved for treatment of mild to moderate Alzheimer’s Disease.
- Rivastigmine: antagonism of butyrylcholinesterase provides greater benefit
- Galantamine: nicotinic receptor agonist
- <u>Adverse effects</u>: muscarinic agonist actions (SLUDGE, bradycardia, bronchospasm) increased adverse effects with CYP450 inhibitors.



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Pharmacokinetics of Donepezil, Rivastigmine, and Galantamine

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DRUG	BIOAVAILABILITY (%)	t (h)	ELIMINATION HALF-LIFE (h)	HEPATIC METABOLISM	REVERSIBLE INHIBITION OF AChE	OTHER CHOLINOMIMETIC EFFECTS
Donepezil	100	3–5	60–90	Yes	Yes	
Rivastigmine	40	0.8–1.8	2	No	No*	BuChEI
Galantamine	85–100	0.5–1.5	5–8	Yes	Yes	nAChR agonist

t_{max} = time to maximum plasma concentration;
*Rivastigmine is a "pseudo-irreversible" inhibitor of AChE and BuChE; AChE = acetylcholinesterase; BuChEI = butyrylcholinesterase inhibitor; nAChR agonist = non-potentiating ligand of the nicotinic receptor.

Referenced from: Golan 2nd edition-Lippincott- For teaching purposes only not for publication.

Memantine

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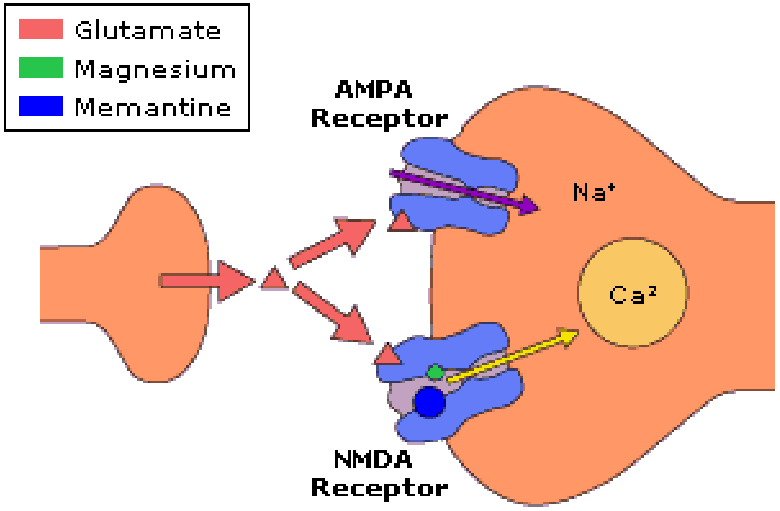
- NMDA Receptor Antagonist; uncompetitive.
- Reduces calcium conductance in pathways of learning and memory. Overactivity causes cell damage and loss of function.
- Indicated for moderate/severe AD.
- <u>Adverse effects:</u> N/V, diarrhea, reduce dose in renal impairment



Aducanumab, w/d 2024

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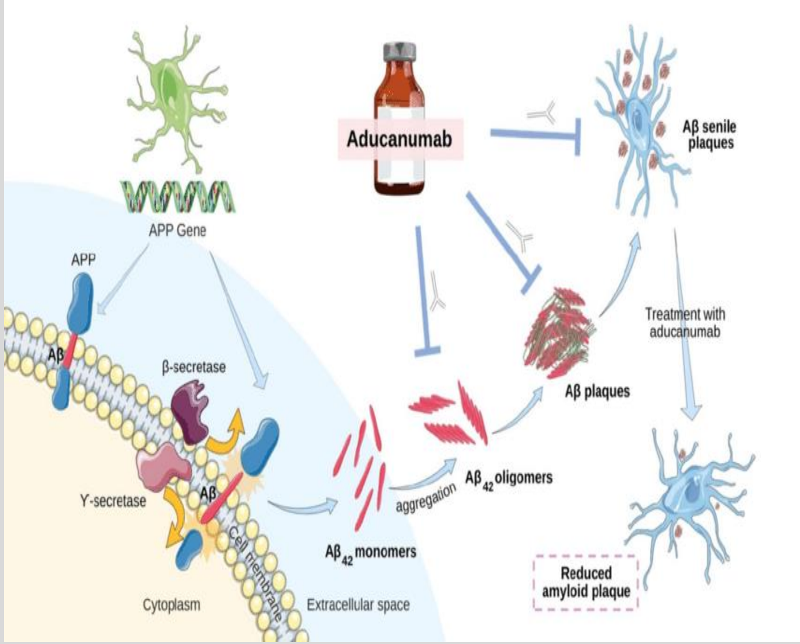
- Monoclonal antibody; targets amyloid beta; Decreases amyloid plaques. Start during mild AD symptoms. PET scan/CSF confirmation
- <u>Adverse effects</u>: Amyloid related imaging abnormalities (ARIA)
  - ARIA-E (edema)
  - ARIA-H (bleeding)
- MRI to be performed specific intervals to infusion



Lecanemab, 2023 and Donanemab, 2024

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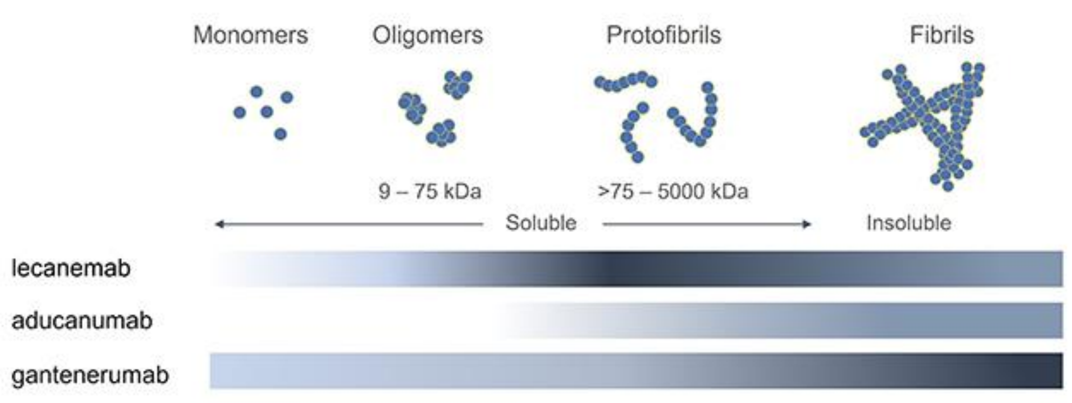
- IV every 2 weeks for lecanemab, monthly for donanemab
- All similar MOA, Lecanemab targets protofibrils
- AE: ARIAs, headache, infusion reactions.
- MRI to be performed at specific intervals



Alzheimer's Disease: Other Drugs

- Atypical Antipsychotics (risperidone, olanzapine, and quetiapine) for agitation and psychosis. Adverse effects?
- SSRIs (depression). Citalopram preferred?
- Antioxidants (Ginkgo Biloba, vitamin E)?
- Caution with antimuscarinics in elderly (Beer's criteria)
- Concurrent geriatric conditions (hypertension, type II diabetes, etc)
- Anticoagulants
- Antimicrobials

Summary Slide

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- **Describe the drugs used to manage Parkinson's disease. Include adverse effects. Explain other drugs associated with Parkinsonism.**
- **Describe the drugs used to manage Alzheimer's disease. Describe additional medications used to manage this population of patients.**

Core References:

1. Katzung & Venderah Basic and Clinical Pharmacology, 16e, 2024; Chapter 28 Pharmacologic Management of Parkinsonism and Other Movement Disorders; Chapter 7 Basic Pharmacology of the indirect-acting cholinomimetics.
2. Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e, 2023; Chapter 21: Treatment of Central Nervous System Degenerative Disorders
3. Drugs for Parkinson's and Alzheimer's disease summary (Maria A. Pino, Ph.D)

Lecture Feedback Form:

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https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R